HAB System Cost Report

GCP Pricing (us-central1)

- **vCPU:** $0.000024 per vCPU-second
- **Memory:** $0.0000025 per GB-second
- **1 Month Runtime:** 2,592,000 seconds
- **Request-Based Estimate:** 20% average utilization ≈ 518,400 seconds/month

Service Configurations and Monthly Cost Estimates (Request-Based)

1. User Service (Backend 1)

- Instances: 1 (always running)
- vCPUs: 2
- Memory: 0.5 GB
- Cost:
  - vCPU: 2 × $0.000024 × 518,400 = $24.88
  - Memory: 0.5 × $0.0000025 × 518,400 = $0.65
  - Total: $25.53/month

2. Prediction Service (Backend 2)

- Instances: 1 (scales on demand)
- vCPUs: 4
- Memory: 2 GB
- Cost:
  - vCPU: 4 × $0.000024 × 518,400 = $49.77
  - Memory: 2 × $0.0000025 × 518,400 = $2.59
  - Total: $52.36/month

3. Frontend Service

- Instances: 1 (always running)
- vCPUs: 2
- Memory: 1 GB
- Cost:
  - vCPU: 2 × $0.000024 × 518,400 = $24.88
  - Memory: 1 × $0.0000025 × 518,400 = $1.30
  - Total: $26.18/month

Total Monthly Cost Summary (Request-Based, 1 Instance Each)

- User Service: $25.53
- Prediction Service: $52.36
- Frontend: $26.18
- Grand Total: $104.07/month

Note: These estimates assume 20% utilization time under request-based billing. Costs will increase if services are accessed more frequently or have more instances.

Regional Pricing Comparison (per instance, always-on)

Google Cloud regions are divided into Tier 1 and Tier 2 based on infrastructure cost and demand. Tier 1 regions offer standard pricing, while Tier 2 regions typically have higher prices due to operational overhead and limited regional availability.

Region	Tier	vCPU/sec (USD)	Mem/sec (USD)	User Service (USD)	Prediction Service (USD)	Frontend (USD)	Total/ month (USD)
us-central1 (Iowa)	Tier 1	0.000024	0.0000025	$25.53	$52.36	$26.18	$104.07
europe-west1 (Belgium)	Tier 1	0.000024	0.0000025	$25.53	$52.36	$26.18	$104.07
europe-west2 (London)	Tier 2	0.0000264	0.00000275	$28.02	$57.50	$28.75	$114.27
europe-west3 (Frankfurt)	Tier 2	0.0000258	0.00000269	$27.37	$56.20	$28.09	$111.66
europe-west4 (Netherlands)	Tier 2	0.0000257	0.00000268	$27.26	$56.00	$27.99	$111.25
Europe-north1 (Finland)	Tier 1	0.000024	0.0000025	$25.53	$52.36	$26.18	$104.07
europe-west6 (Zurich)	Tier 2	0.0000275	0.00000285	$29.18	$59.87	$30.07	$119.12

AWS Equivalent Pricing Comparison (Lambda + ECR)

AWS Lambda (us-east-1 region used for comparison):

- vCPU (1ms = 1/1000 sec): $0.0000166667 per ms for 2 vCPU equivalent
- Memory: $0.000000208 per MB-ms = $0.000208 per GB-sec

Estimated Monthly Usage: 518,400 seconds (same as GCP)

1. User Service (Backend 1)

- Memory: 512 MB = 0.512 GB
- Cost: 0.512 GB × $0.000208 × 518,400 = $55.37

2. Prediction Service (Backend 2)

- Memory: 2 GB
- Cost: 2 GB × $0.000208 × 518,400 = $215.04

3. Frontend Service

- Memory: 1 GB
- Cost: 1 GB × $0.000208 × 518,400 = $107.52

AWS Total Estimate

- User Service: $55.37
- Prediction Service: $215.04

- Frontend: $107.52
- Grand Total (Lambda-based): $377.93/month

Summary: GCP vs AWS

Platform	Total Monthly Cost
GCP	$104.07
AWS	$377.93

Why GCP Was Chosen

- **Pricing Advantage:** GCP Cloud Run is significantly more affordable for container-based apps with moderate, predictable traffic.
- **Easier Container Deployment:** No need to manage separate container services; Cloud Run directly runs Docker images.
- **Always-On Option:** GCP offers min instances (e.g., for frontend/backend), which Lambda does not support.
- **Better concurrency per instance:** Cloud Run supports up to 80 concurrent requests per container instance.
- **Tighter Integration:** GCP services like Artifact Registry integrate more smoothly with Cloud Run for ML workflows.